

Choosing an AI Agent Framework & LangChain Fundamentals

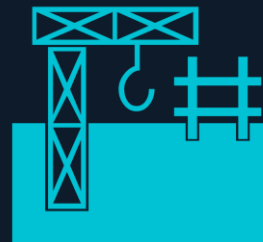
- Understanding AI Agent Frameworks
- Learning LangChain Architecture
- Introduction to Chains & Workflows



AI FRAMEWORK

What is a

Framework?



FRAMEWORK
Structure • Tools • Rules

■ Pre-Built Structure

Ready-made components so you don't start from scratch

✂ Tools & Guidelines

Built-in tools, best practices, and development rules

♻ Reusable Components

Use proven code blocks across multiple projects

★ *Instead of writing everything manually, we reuse proven components!*

What is it?

A platform to build intelligent AI systems

LLM

Language Models (LLMs)

The reasoning engine

API

Tools & APIs

Connect to external services

MEM

Memory

Stores conversation history

DT

Decision Making

Autonomous action selection

Are Needed

AB

Abstraction

Hides complex AI logic so you focus on features, not internals.

SC

Scalability

Supports large, real-world production applications seamlessly.

IN

Integration

Connects external systems — APIs, databases, and tools.



POWER UP
Your Development

⚠ Without frameworks...

- Slow development cycles
- Hard to maintain code
- Difficult to scale up
- Complex integrations

POPULAR TOOLS

AI Agent Frameworks



LangChain

Flexible, modular framework for building LLM-powered apps. Best for custom, complex AI workflows.



Flowise AI

Drag-and-drop visual interface. No-code builder for rapid prototyping.



AutoGen

Microsoft's framework for orchestrating multiple AI agents to solve complex tasks.

Choose based on your project's complexity and team's skill level

FOR STUDENTS

Learning Goals

1

Understand AI App Building

How intelligent applications are designed from the ground up.

2

Connect the Components

How LLMs, memory, tools, and agents work together.

3

Build Smarter Systems

Create AI systems with less effort using abstractions.

4

Think System Design

Understand architecture and workflow design, not just coding.



LEARN • BUILD • GROW

Key Factors for Choosing a Framework

- **Ease of Use**

Is it beginner-friendly?

- **Integration**

API & tool support?

- **Customization**

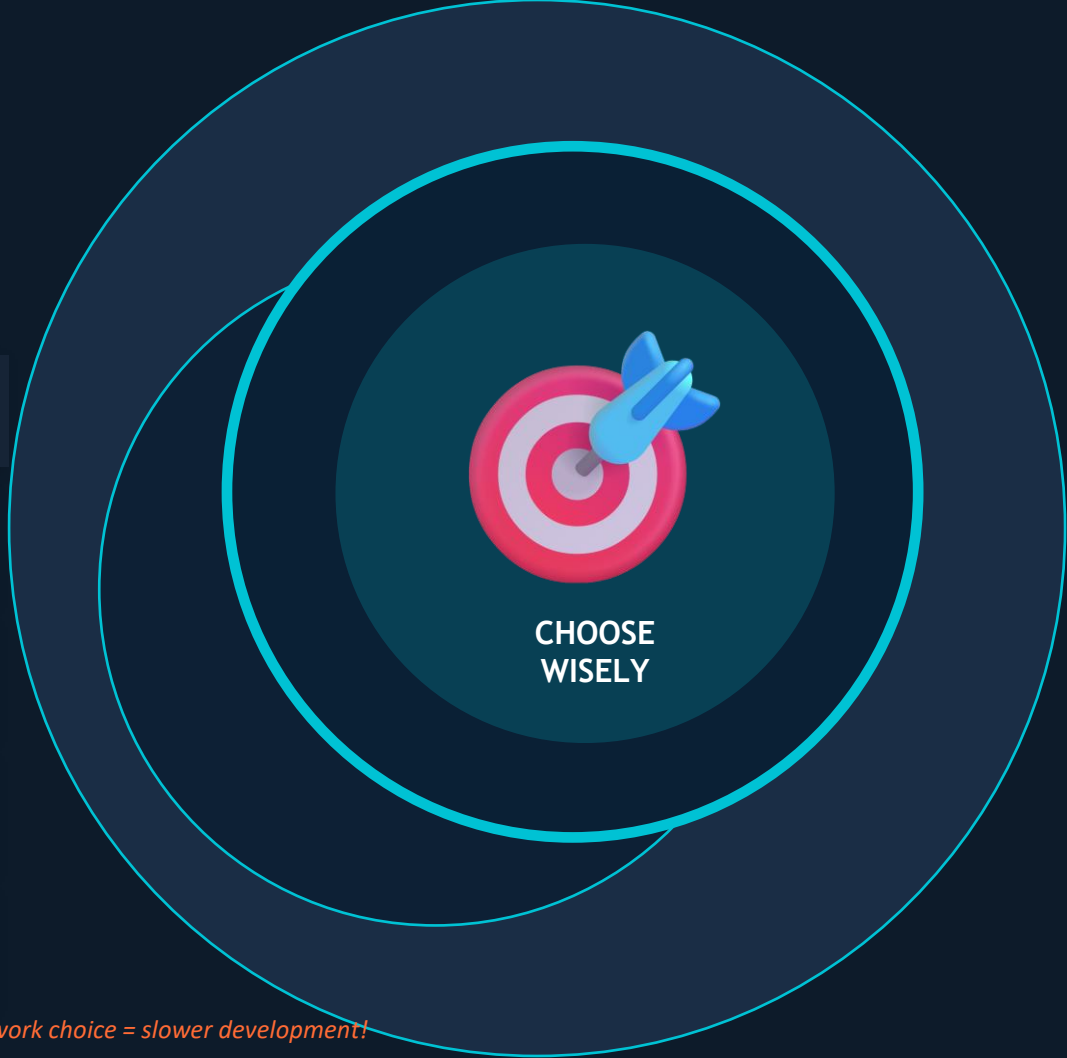
Can we modify behavior?

- **Performance**

Fast and efficient?

- **Community**

Resources & documentation?



Wrong framework choice = slower development!

Which Framework Should You Choose?



STARTUP MODE

- Faster development cycles
- Visual drag-and-drop interface
- No deep coding required
- Launch products quickly

VS



ENTERPRISE

- Full control and flexibility
- Scalable for large systems
- Custom integrations
- Production-grade pipelines

Choice depends on project complexity, team size & time constraints

LANGCHAIN

What is LangChain?

A framework designed for LLM-based applications



Acts as bridge between LLMs and real-world tools



Build chatbots, agents, automation systems



Connects different AI components into workflows



Supports 50+ integrations with APIs and databases



Architecture & Components

LM

LLM

The brain — generates responses and reasoning

PT

Prompt Templates

Structures input text for consistent AI output

CH

Chains

Defines the step-by-step workflow execution

TL

Tools

Perform external actions — APIs, search, compute

ME

Memory

Stores past interactions for context awareness

AG

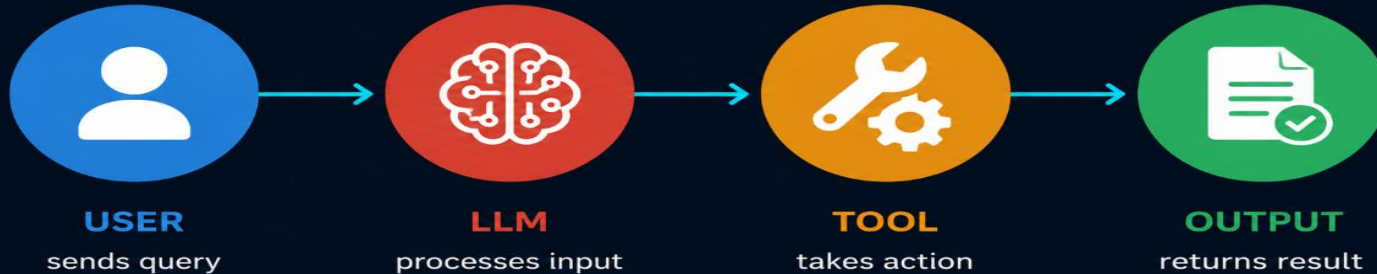
Agents

Decision-makers that choose actions dynamically

CONCEPT

LangChain in Action

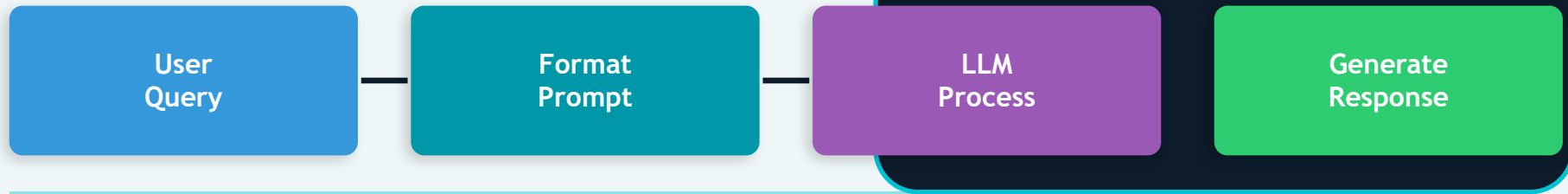
LangChain = Multiple AI Components + Intelligent Workflow



Real Example: User asks weather → system processes → API called → result shown

What is a Chain?

A sequence of steps executed one after another, each step processes and passes data forward.



★ Why This Matters:

- Breaks complex problems into manageable, testable steps
- Ensures structured, consistent execution every time
- Improves accuracy by isolating each processing stage

Types of Chains in LangChain



Simple Chain

One-step execution. Input goes in, output comes out. Perfect for straightforward tasks.

Example Flow:

Query → LLM → Answer

Best for: Single queries, Q&A



Sequential Chain

Multiple steps executed in order. Output of step 1 becomes input for step 2.

Example Flow:

Input → Step1 → Step2 → Step3 → Output

Best for: Multi-step reasoning



Router Chain

Selects different paths based on input type. Intelligent routing to specialist chains.

Example Flow:

Input → Router → [Chain A / Chain B]

Best for: Multi-domain systems

Step-by-Step Example Workflow



01

STEP 01 – User Input

User types: "What is the weather in Chennai today?"

02

STEP 02 – Input Processing

Prompt is formatted with context and instructions for the LLM

03

STEP 03 – LLM Generates Response

AI model reasons and decides to call the weather API tool

04

STEP 04 – Output Returned

Formatted weather result displayed in natural language

Limitations & Challenges



Hard to Debug

Complex workflows with many steps become difficult to trace and debug when errors occur.



Integration Failures

Tool and API integrations can fail — network issues, rate limits, schema changes.



Performance Issues

Multiple sequential steps increase latency — each tool call adds response time.



Learning Curve

Requires time to understand the ecosystem, architecture, and best practices.

Understanding limitations helps you build smarter, more resilient AI systems!